Bioinformatics & Algorithms for Bioinformatics
25/26

Lab06 - BLAST

1. The BLAST programs (Basic Local Alignment Search Tool) are very efficient for comparing biological sequences, such as DNA or protein sequences, with databases of known sequences. The most commonly used are BLASTN for nucleotides and BLASTP for proteins (https://blast.ncbi.nlm.nih.gov/Blast.cgi).

Note that depending on BLAST server usage, some results may take longer to appear (a few seconds to a few minutes). Please be patient!

a. First, consider BLASTN. Copy the following nucleotide sequence and enter it in "Enter Query Sequence". In "Job Title", write "Unknown DNA Sequence". In "Database", you can leave the default values ("Standard Databases > Nucleotide collection (nr/nt)"). In "Program Selection", leave "Highly Similar Sequences" and click on "BLAST". Your results should be displayed shortly, in a list with different views: *Descriptions*, *Graphic Summary*, *Alignments*, *Taxonomy*.

   Based on the results obtained, which organism does this sequence come from? And which gene is it from?

```
ATGGGTAAGGAGGACAAGACTCACCTTAACGTCGTCGTCATCGGCCACGTCGACTCTGGCAAGTCGACCACTGTAAGT
ACAACCAACAGCGGGTTGCTTATCTGCACTCGGAATCCGCCAAACCTGGCAGGGTATCACCAAAACATCTTGCTAACT
TTTGACAGACCGGTCACTTGATCTACCAGTGCGGTGGTATCGACAAGCGAACCATCGAGAAGTTCGAGAAGGTTAGTC
AATATCCCTTCGATTACGCGCGCTCCCATCGATTCCCACGATTCGCTCCCTCACTCGAAACACATCCATTACCCCGCT
CGAGTCCGAAAATTTTGCGGTGCGACCGTGATTTTTTCTGGTGGGGTATCTTACCCCGCCACTCGAGTCACGGATGCG
CTTGCCCTGTTCCCACAAAACCTTACCACCCTGTCGCGCACTACATGTCTTGCAGTCACTAACCACTGGACAATAGGA
AGCCGCCGAGCTCGGAAAGGGTTCCTTCAAGTACGCCTGGGTTCTTGACAAGCTCAAAGCCGAGCGTGAGCGTGGTAT
CACCATTGATATCGCTCTCTGGAAGTTCGAGACTCCTCGCTACTATGTCACCGTCATTGGTATGTTGTCACCGTCTCA
CACTATCATGTATTCATCATGCTAACATCTCTCTCAGATGCCCCCGGTCATCGTGATTTCATCAAGAACATGATC
```

b. Analyze the following sequence and identify the organism and gene it belongs to:

```
ATGGGAAAGGAGAAGACCCACATCAACATCGTTGTCATTGGGCACGTAGATTCAGGGAAGTCTACCACGACTGGCCATCTGA
TCTATAAATGTGGCGGGATCGACAAGAGAACAATTGAAAAGTTCGAGAAGGAGGCTGCCGAGATGGGAAAGGGCTCCTTCAA
ATATGCCTGGGTCTTGGACAAACTTAAAGCTGAACGTGAGCGTGGTATCACCATTGATATCTCCCTGTGGAAATTTGAGACC
AGCAAGTACTATGTTACCATCATTGATGCCCCAGGACACAGAGACTTCATCAAAAACATGATTACAGGCACATCCCAGGCTG
ACTGTGCTGTCCTGATCGTTGCTGCTGGTGTTGGTGAATTTGAAGCCGGTATCTCCAAGAACGGGCAGACCCGTGAGCATGC
CCTTTTGGCTTACACCCTGGGTGTGAAACAACTAATTGTTGGCGTTAACAAAATGGATTCCACTGAGCCACCCTATAGCCAG
AAGAGATACGAAGAAATTGTTAAGGAAGTCAGCACCTATATTAAGAAAATTGGCTACAACCCCGACACAGTAGCATTTGTGC
CAATTTCTGGCTGGAATGGTGACAACATGCTAGAACCAAGTGCTAATATGCCATGGTTCAAGGGATGGAAAGTCACCCGTAA
GGACGGCAATGCCAGTGGAACCACCCTGCTTGAAGCTCTGGATTGCATTCTGCCACCAACTCGCCCAACTGACAAACCCTTG
CGTTTGCCTCTCCAGGATGTCTATAAAATTGGTGGTATTGGTACTGTCCCTGTGGGTCGTGTGGAGACTGGTGTTCTCAAAC
CTGGCATGGTGGTCACCTTTGCTCCAGTCAATGTAACAACTGAAGTGAAGTCTGTAGAAATGCACCATGAAGCATTGAGTGA
AGCCCTTCCTGGGGACAATGTGGGCTTTAATGTCAAAAACGTGTCTGTCAAAGATGTCCGTCGTGGCAATGTGGCTGGTGAC
AGCAAAAATGATCCACCCATGGAAGCTGCTGGCTTCACAGCTCAGGTGATTATTTTGAACCATCCAGGCCAAATCAGTGCTG
GATATGCACCTGTGCTGGATTGTCACACAGCTCACATTGCTTGCAAGTTTGCTGAGCTGAAGGAGAAGATTGATCGTCGTTC
TGGGAAAAAGCTGGAAGATGGCCCTAAATTCTTGAAATCTGGTGACGCTGCCATCGTTGATATGGTTCCTGGCAAGCCCATG
TGTGTCGAGAGCTTCTCTGATTATCCTCCCCTGGGCCGTTTTGCTGTGCGTGACATGAGACAGACAGTCGCTGTGGGTGTCA
TCAAAGCAGTGGACAAGAAGGCAGCTGGAGCTGGCAAGGTCACCAAGTCTGCCCAGAAAGCTCAGAAGGCTAAATGA
```

c. Now let's compare the two sequences found. Return to the main BLASTN interface and enter the first sequence in "Enter Query Sequence". Then, click on "Align two or more sequences" and copy the second sequence into the box that appears below. Keep the default Program Selection as "Highly similar sequences". What happens?

d. Let's try a more sensitive algorithm. Click on "Edit Search" and choose "More dissimilar sequences (discontiguous megablast)".
   i. Investigate the alignment of the sequences under "Alignments". Try changing the Alignment View between "Pairwise" and "Pairwise with dots for identities". In how many ranges are the sequences aligned?
   ii. Can you obtain similar results using a Dotplot? Consider using Dotlet (https://dotlet.vital-it.ch) and indicate the parameters used.

e. Consider now the human AQP3 gene (*aquaporin 3*). You can extract this sequence from the NCBI gene database (https://www.ncbi.nlm.nih.gov).
   Copy the entire sequence into BLASTN and search for "Highly Similar Sequences" (don't forget to uncheck the "Align two or more sequences" box).
   i. Indicate the description of the first 3 matches. What do the **E-value** and **Perc. Identity** mean?
   ii. Check the alignment of the last match (*Muscardinus avellanarius genome assembly, chromosome: 13*) and interpret your results. What do the characters " " (space), "|", and "-" indicate?
   iii. Check the "Distance tree of results". What can you observe? In which other species can the AQP3 sequence also be found?

Search results

Items: 1 to 20 of 735 << First < Prev Page 1 of 37 Next > Last >>

See also 7 discontinued or replaced items.

Name/Gene ID	Description	Location	Aliases	MIM
AQP3 ID: 360	aquaporin 3 (Gill blood group) [*Homo sapiens* (human)]	Chromosome 9, NC_000009.12 (33441160..33447593, complement)	AQP-3, GIL	600170
Aqp3 ID: 11828	aquaporin 3 [*Mus musculus* (house mouse)]	Chromosome 4, NC_000070.7 (41092724..41098183, complement)	AQP-2	

f. Now focus on BLASTP to analyze proteins. Consider the SR1_HUMAN (search for the sequence on NCBI or UniProt). Copy the sequence into "Enter Query Sequence" and keep the options ("Database: Standard databases > Non-redundant protein sequences (nr)"). In "Organism", select "Homo sapiens (taxid:9606)". Keep the program "blastp (protein-protein BLAST)".
   i. Indicate the **Query Cover**, **E-value**, and **Perc. Identity** of the first 3 matches.
   ii. Check the "Alignments" of the first 3 matches. While the first 2 matches have an Identity of 100%, the third match has 99%. Indicate the **Query** and the **Sbjct** where a substitution occurred (i.e., where the sequences are not exactly identical).
   iii. Explore the alignment of the fourth match (68.61% identity) and interpret your results. What do the characters " " (space), "+", and "-" indicate?

2. For the amino-acid sequences of the **protein HBA Human (P69905)** and the **transcript sequence of the HBB Human gene (Refseq NM_000518.5)**, search independently the most similar **protein** sequences in a database of non-redundant protein sequences database (nr).
    a. Select the most appropriate BLAST program for the effect.
    b. What is the E-value of the Top-hit from a species different from human?
    c. For each case, select the top 10 protein sequences from 10 different species (rather than human).

3. Use **BioPython** (https://biopython.org) to implement a program that retrieves the information asked in the previous exercise. You can have the sequence files locally stpres and read them directly in the script for processing. Add a print function that outputs a "pretty-print" of a table with the following elements: target_sequence, species, length, E-value, and %identity.